



Intriguing Signal

Alien hunters discover strange transmissions coming from the direction of Proxima Centauri. <http://digit.in/jan21-17>



GN-z11

Astronomers have spotted what they deem to be the most distant and oldest galaxy ever discovered. <http://digit.in/jan21-18>

GALAXY EVOLUTION

Galaxies are not static objects, they just change over incredibly long periods of time

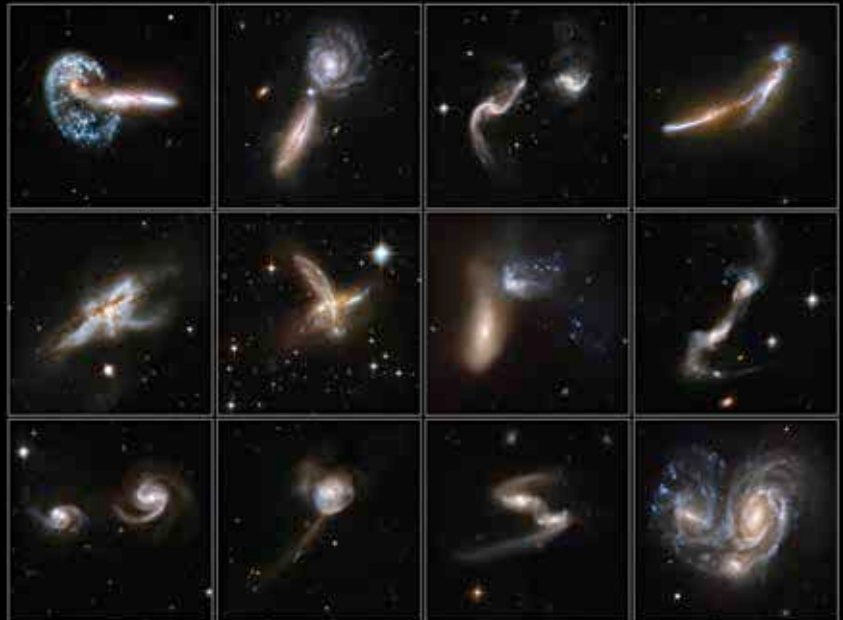
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or most of history, we did not know that galaxies existed. It is only today that scien-

tists think of galaxies as islands of stars embedded in a substrate of dark matter. Till around the 1920s, galaxies were not even distinguished from nebulae. While the ancient astronomers could see some of the nearby galaxies, they just referred to it as clouds of gas. It was only after the discovery of telescopes, and observations of the nearby Andromeda, that scientists began to suspect that local clusters of stars might exist. The concept of a galaxy till then was still the Milky Way as it appears in the night sky. In fact for a short period of time, Andromeda and some other spiral galaxies to be discovered were called "island universes" in the 19th century, but the term fell out of use, and we now have galaxies.

On 26th April 1920, what is now known as the Great Debate took place between two scientists, Harlow Shapley and Heber Curtis at the Smithsonian Museum of Natural History. Shapely maintained that the other "spiral



A gallery of galaxy interactions captured by the HST

nebulae" to be observed were part of the Milky Way galaxies, while Curtis argued that they were independent galaxies at incredible distances. Apart from the nature of the galaxies, the debate was also about the size of the universe. After the debate the scientific papers submitted by both the scientists were up for scrutiny. Edwin Hubble went on to settle the matter, by demonstrating that Andromeda was indeed an independent galaxy. Edwin Hubble's observation made the universe much, much bigger than few imagined it to be. Since the 20th century, astronomical instruments have been finding galaxies faster than scientists could study them. In 1995 the Hubble Space Telescope took an unusually long exposure of a dark portion of the

sky, and found over 3,000 objects, all of which were galaxies. The Hubble Deep Field image allowed astronomers to estimate that there are about 125 billion galaxies in the observable universe.

GALACTIC DYNAMICS

The most apparent differences between all the observed galaxies is the shape, which led to a classification scheme. Hubble came up with the scheme in 1926, and it continues to be used today. They are called elliptical, lenticular, spiral and irregular galaxies. From left to right across the scale, the elliptical galaxies get increasingly elongated. The Milky Way is a barred spiral galaxy. The lenticular galaxies are made up of a central bulge and disc like components, with the classification integers

Credit: ESA